

NEUROMECHANISMS OF LONG-TERM MEMORY RETRIEVAL

POOPA KAEWBUAPAN

A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR
THE DEGREE OF BACHELOR OF SCIENCE (COMPUTER SCIENCE)
SCHOOL OF INFORMATION TECHNOLOGY
KING MONGKUT'S UNIVERSITY OF TECHNOLOGY THONBURI

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Poopa Kaewbuapan

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Academic Year 2024

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Abstract

Forgetting often occurs due to failures in long-term memory (LTM) retrieval, even when the memory content itself remains intact. Contemporary models of human memory introduces a short-term, or working memory (WM), component that acts as a buffer to LTM. WM performance is known to be influenced by "attention" neuromechanism that selectively filter information necessary for both remembering and later recalling. This raises an intriguing question: is LTM retrieval also modulated by attention, or does it rely on an entirely different neuromechanism? To address this, we devised two behavioral experiments to measure and compare WM and LTM retrieval performance while recalling one or two objects under varying attentional demands. To further identify the neuromechanism behind LTM retrieval, we modeled both WM and LTM with a mixture model to extract two parameters: mnemonic precision (SD) and guessing rate (g), which reflect the utilization of attention and decision-making neuromechanism, respectively. We discovered reversal of trends in mixture model parameters of LTM retrieval data when compared to WM, supporting decision-making as neuromechanism that modulated LTM retrieval.

Keywords : Neuromechanism, Long-Term Memory Retrieval, Working Memory, Divided Attention, Decision-Making

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List of Symbols

$^{\circ}$	Degrees (angular measurement)
$\pm$	Plus-minus
α	Alpha (significance level)
μ	Mu (population mean)
σ	Sigma (standard deviation)
p	P-value
F	F-statistic - measures the ratio of variance between groups to variance within groups, with larger values indicating stronger evidence of group differences
t	T-statistic - measures the difference between groups in standard error units, with negative values indicating the first group's mean is lower than the second group's mean.
n	Sample size

Vocabularies and Abbreviations

WM	Working Memory - temporary active representation and manipulation of information
LTM	Long-term Memory - storage system for later retrieval of information
SD	Memory Precision parameter measured in degrees of visual angle from target location
g	Guessing Rate parameter indicating probability of random responses
Response Error	Error distance from answered location relative to the target position measured in degrees
Reaction Time	Time from stimulus onset until response
Continuous Report Task	Psychological task where participant has to report a certain value out of range of continuous values
Mixture Model	Computational model combining Gaussian and uniform distributions to characterize memory responses
Swap Model	Extended mixture model that includes additional parameter for binding errors in multi-item memory
IRB	Institutional Review Board for human subject research approval
B	Swap Model parameter representing probability of choosing distractor item
SE	Standard Error
Hz	Hertz - unit for measuring frequency

Chapter 1

Introduction

1.1 Background and Rationale

Human memory consists of multiple interacting systems that enable us to encode, store, and retrieve information. A key distinction exists between working memory (WM), which temporarily maintains information for ongoing tasks, and long-term memory (LTM), which stores information for extended periods. Contemporary models propose that WM acts as a buffer through which information from LTM must pass [1] [8].

While WM is known to be modulated by attention mechanisms that filter and select relevant information [7], the neuromechanisms governing LTM retrieval remain debated. Both neuroimaging and electrophysiology studies have demonstrated that attention-related neural signals predict WM performance [15] [13]. However, whether similar attentional mechanisms support LTM retrieval, or if LTM relies on distinct neuromechanisms remains unknown. Understanding these mechanisms is crucial for both visual memory research and applications in education and cognitive rehabilitation.

Recent studies examining the precision of visual working memory have revealed that memory resolution decreases continuously as more items are stored [2] [18]. Additionally, alpha-band oscillations (8-12 Hz) have been shown to track both the maintenance of spatial information in working memory [5] and the retrieval of precise spatial locations from long-term memory [16], suggesting possible shared mechanisms between working memory maintenance and long-term memory retrieval.

1.2 Motivation

The relationship between attention and memory systems has significant implications for understanding both normal cognitive function and memory-related disorders. Current research suggests that working memory acts as an interface between attention and long-term storage, but the precise mechanisms remain unclear [3]. By investigating how attention modulates both working memory and long-term memory retrieval, we can better understand the fundamental organization of human memory systems and potentially develop more effective interventions for memory enhancement.

1.3 Research Questions

The research questions addressed in this study are as follows:

1. How do attentional demands affect working memory versus long-term memory retrieval performance?
2. What are the specific neuromechanisms that support long-term memory retrieval?

1.4 Objectives

The objectives of this research can be summarized as follows:

1. To dissociate the roles of attention and decision-making in long-term memory retrieval through behavioral experiments
2. To compare working memory and long-term memory performance under varying cognitive loads
3. To identify specific neuromechanisms underlying long-term memory retrieval through computational modeling

1.5 Expected Benefits

According to the objective of this research, we expect benefits as follow:

1. A deeper understanding of the neuromechanisms that govern long-term memory retrieval, which can inform both basic research and clinical applications

1.6 Scope of Study

This study investigates the neuromechanisms of long-term memory retrieval using non-invasive behavioral methods. The research focuses on healthy adult participants aged 18-35 years, employing controlled experimental paradigms that manipulate attentional demands and memory load. We utilize established computational models to characterize memory performance and extract key parameters reflecting different cognitive processes. The study employs PsychToolbox [10] for stimulus presentation and implements n-way repeated measures ANOVA and paired t-test statistical analyses to test our hypotheses about memory mechanisms.

1.7 Thesis Organization

This thesis is divided into five chapters as follows:

- Chapter 1 is the introduction of the study. This chapter describes the background of memory systems, research motivation, objectives, and scope of investigating neuromechanisms underlying long-term memory retrieval.
- Chapter 2 presents the literature review. This chapter reviews previous studies examining relationships between working memory, attention, and long-term memory retrieval.
- Chapter 3 proposes the research methodology. This chapter illustrates the experimental design and procedures, including participant recruitment, behavioral tasks, data collection, and analytical approaches using computational modeling and statistical tests.
- Chapter 4 describes the results and discussion. This chapter presents behavioral and modeling findings from both working memory and long-term memory experiments, analyzing how different attentional demands affect working memory and long-term memory performance.

- Chapter 5 explains the conclusions and future works. This chapter summarizes key findings about neuromechanisms of long-term memory retrieval, discusses contributions, and suggests directions for future research.

Chapter 2

Literature Reviews

2.1 Challenges It Faces Indicating The Need For This Research

Contemporary models of human memory propose that working memory (WM) acts as a buffer through which information from long-term memory (LTM) must pass [1] [8]. While we understand that WM is modulated by attention mechanisms that filter and select relevant information [7], the neuromechanisms governing LTM retrieval remain debated.

Previous studies examining memory mechanisms have focused primarily on WM, showing that memory precision decreases as more items are stored [18] [2]. Neuroimaging and electrophysiological studies demonstrate that attention-related neural signals predict WM performance [15] [13]. However, whether similar attentional mechanisms support LTM retrieval, or if LTM relies on distinct neuromechanisms, remains unknown.

Recent work suggests the possible involvement of decision-making processes in memory. For instance, perceptual bias studies show that longer response times allow for better decision-making in visual discrimination tasks [4]. This raises the possibility that LTM retrieval might depend more on decision-making mechanisms than attentional control.

The classical WM model proposes a limited capacity of 3-4 independent "slots" for storing visual information [12] [3]. However, newer models like the mixture model [18] and swap model [2] suggest more complex mechanisms involving both mnemonic precision and guessing parameters. These models provide tools for dissociating different cognitive mechanisms but have not been extensively applied to understanding LTM retrieval.

2.1.1 Conclusion

There are several key challenges in understanding LTM retrieval mechanisms:

1. While WM's reliance on attention is well-established, the neuromechanisms underlying LTM retrieval remain unclear
2. Existing studies focus primarily on WM, leaving a gap in our understanding of LTM retrieval processes

These challenges highlight the need for systematic investigation comparing how attention and decision-making mechanisms contribute to both WM and LTM retrieval. Understanding these mechanisms could improve our theoretical models of memory and potentially lead to better interventions for memory enhancement [6].

Chapter 3

Research Methodology

This chapter describes the experimental design, procedures, and analysis methods used to investigate neuromechanisms underlying long-term memory retrieval. The research consists of two main experiments examining working memory and long-term memory performance under varying cognitive demands.

3.1 Participant Recruitment and Data Collection Setting

The study recruited 17 participants through convenience sampling from King Mongkut's University of Technology Thonburi undergraduate population, with a mean age of 20.94 ± 1.144 years (age range 18-35 years). All experimental procedures were approved by the institutional review board, and participants provided written informed consent prior to participation. Three participants were excluded from model-based analysis on long-term memory data due to unreliable parameter fits in computational modeling (10 males and 4 females left).

Table 3.1: Participant Demographics

Parameter	Value
Sample Size	17 participants
Male	11 (64.7 %)
Female	6 (29.42 %)
Age Range	18-35 years
Mean Age	20.94 ± 1.144 years

Data collection was conducted at the Neuroscience Center for Research and Innovation (NX), KMUTT, between August and October 2023. Each participant completed four experimental sessions over consecutive days, with Day 1 consisting of a 2-hour working memory assessment and Days 2-4 comprising 3-hour long-term memory sessions. Participants received monetary compensation of 300 bahts per session.

3.2 Experimental Design and Task Structure

3.2.1 Experiment 1: Working Memory Assessment

The working memory assessment conducted on Day 1 employed three distinct conditions to examine how attention modulates working memory performance. In the single stimulus condition, participants viewed one colored dot and reported its location after a brief retention interval using a circular response ring. This condition established baseline memory precision without additional cognitive load. The two stimuli focused attention condition presented two colored dots simultaneously with an early cue indicating the relevant target, requiring participants to selectively

attend to one item while ignoring the other. The two stimuli divided attention condition similarly presented two dots but provided the response cue only after the retention interval, necessitating maintenance of both items in memory.

Stimuli geometry and timing followed established protocols from previous research [5]. We use a 24-inch LCD monitor operating at 144Hz refresh rate for stimuli presentation. Each condition comprised 27 trials with total of 81 trials per block, with brief breaks provided every 10 minutes to minimize fatigue.

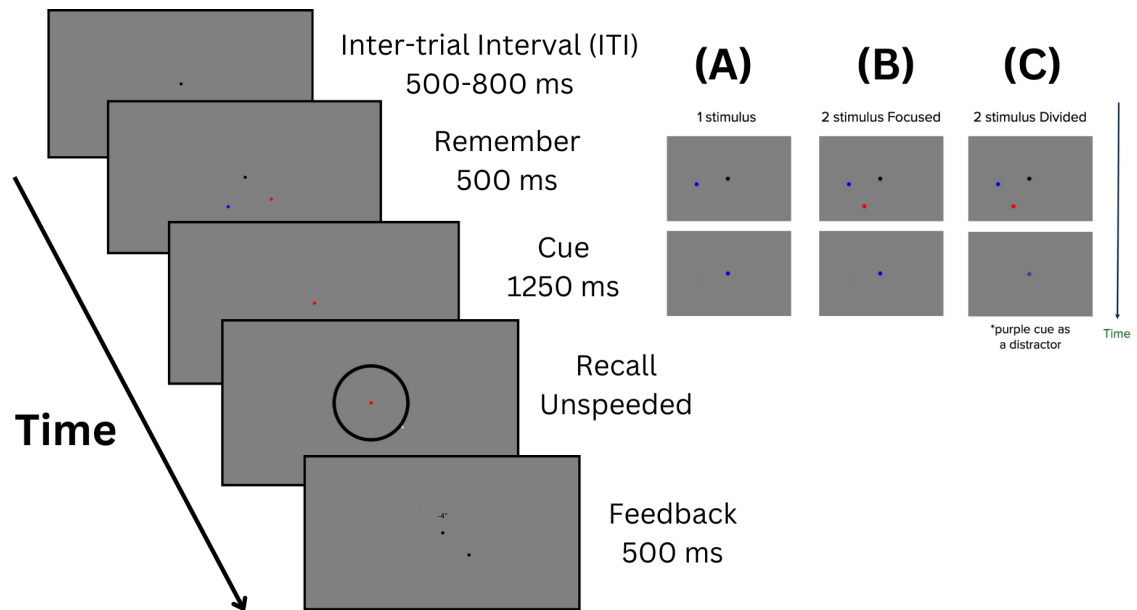


Figure 3.1: Working memory task structure. (A) Single stimulus presentation. (B) Two stimuli with focused attention, showing early cue. (C) Two stimuli with divided attention, showing delayed cue.

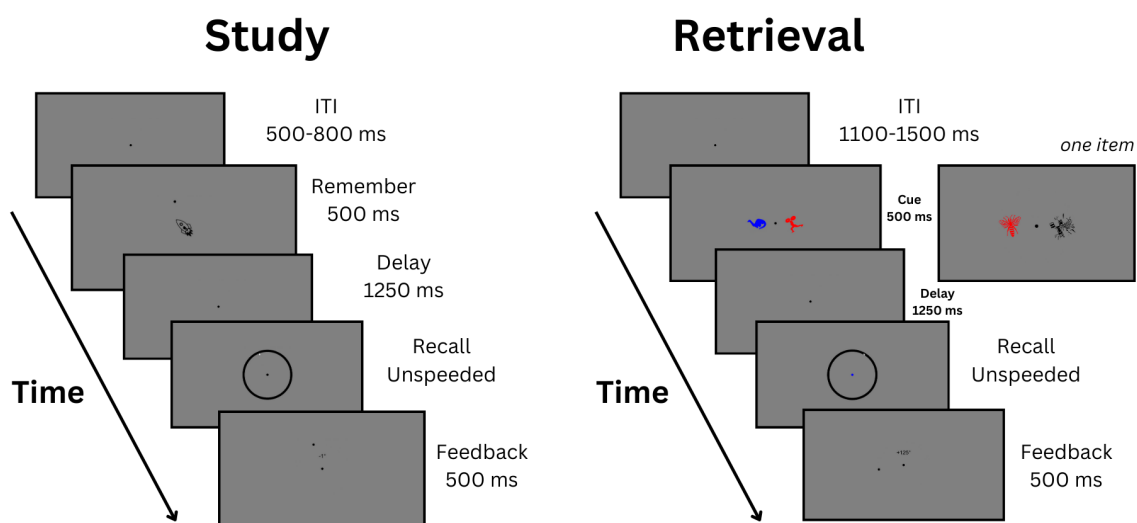


Figure 3.2: Long-term memory task structure. Study block showing sequential presentation of object-location pairs. Retrieval block illustrating the two possible cases: one item cue and two items cue that may be grouped or non-grouped.

3.2.2 Experiment 2: Long-term Memory Assessment

The long-term memory assessment spanned three consecutive daily sessions, each designed to examine memory retrieval under varying attentional demands. During study phases, participants learned 162 unique object-location pairs through sequential presentation to eliminate potential load effect. Objects consisted of 162 cliparts mapped on latent associated locations specified as points on a 360-degree circular response ring. There are 27 possible spatial associates on the response ring, with 9 equidistant spots that are 40 degrees apart, each spot can be jittered to the left or right with 10 degrees deviation to mitigate possible mnemonic exploitation. The cliparts will be randomly tilted at $[-45^\circ \ 45^\circ]$ when being presented to mitigate possible feature exploitation.

The retrieval phase implemented three conditions to dissociate attention and memory mechanisms. The single recall condition presented one target object with a scrambled placeholder to balance sensory inputs on both left and right visual field, providing baseline retrieval performance. The two-item grouped recall condition presented consistent object pairs across blocks to reduce attentional demands in later stage of learning, while the two-item non-grouped recall condition used random pairings. Additionally, individual retrieval blocks tested all objects separately to verify comprehensive learning and establish baseline performance measures.

Stimuli geometry and timing followed established protocols from previous research [16].

3.3 Data Collection and Analysis Methods

3.3.1 Behavioral Measurements

The study collected two primary behavioral measures across all experimental conditions. Response error was calculated as the angular difference between reported and true location, ranging from -180 degrees (counterclockwise) to +180 degrees (clockwise), serving as the primary measure of mnemonic precision. Reaction time measured from stimulus onset until the participant respond with a mouse click.

3.3.2 Statistical Analysis Framework

For the working memory task, two-way repeated measures ANOVA examined the effect of each repetitions and WM conditions (one stimuli, two stimulus focused, two stimulus divided) on response error and one-way repeated measures ANOVA were use for the parameter data, followed by paired t-tests. Long-term memory analysis employed two-way repeated measures ANOVA examining the effects of each sessions and LTM conditions (one item, two items grouped, two items non-grouped) on response error followed by paired t-tests.

3.3.3 Computational Modeling

The study implemented two computational models to characterize memory mechanisms. The mixture model, based on established frameworks [18], extracted two key parameters: SD (memory precision) and g (guessing rate). The swap model extended this framework by adding a binding error parameter (B) to account for non-target responses in multiple-item conditions [2]. Model selection utilized two criteria including Corrected Akaike Information Criterion (AICc) and

Bayesian Information Criterion (BIC), in which lower scores implies model parsimony. Both model selection and model fitting employed through the MemToolbox [14].

All analyses were performed using MATLAB R2023a (MathWorks) with statistical significance set at $\alpha = 0.05$.

Chapter 4

Results and Discussion

This chapter presents analyses of behavioral and computational modeling results from both working memory and long-term memory experiments. Results are organized hierarchically from behavioral performance to model-based analyses.

4.1 Working Memory Task Results

4.1.1 Behavioral Performance

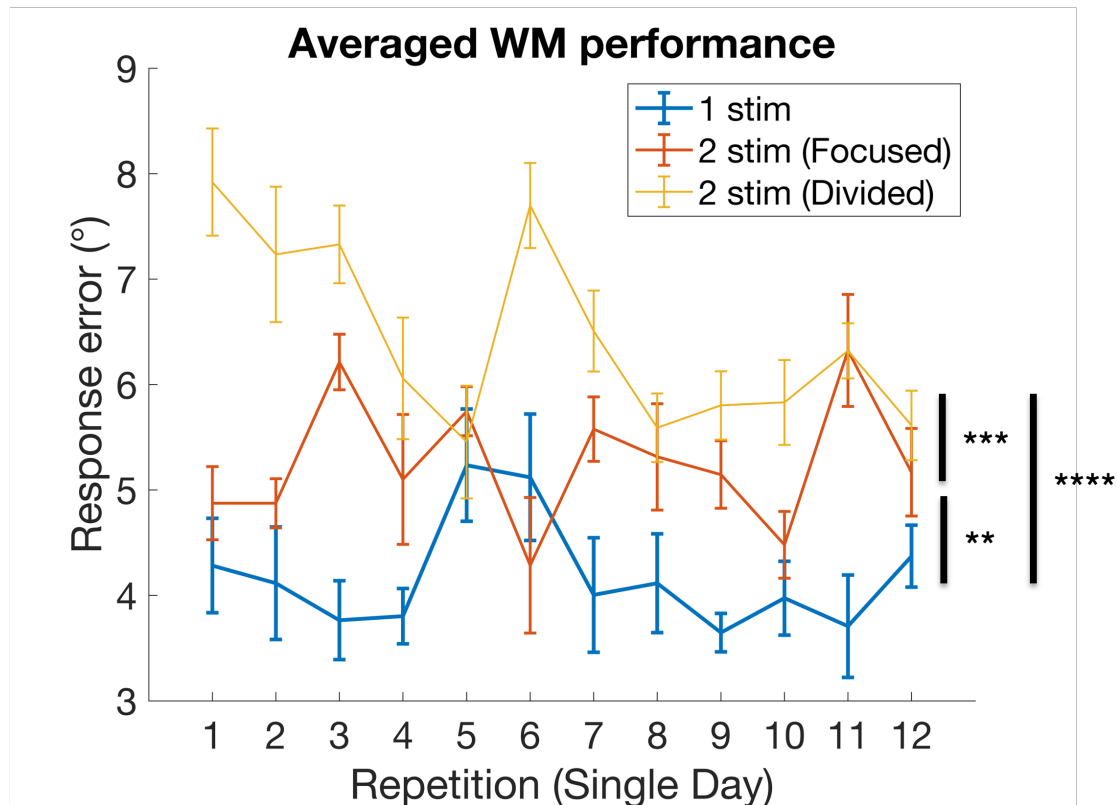


Figure 4.1: Averaged working memory performance across conditions for all 17 participants showing significant increases in response error with increased attentional load. Error bars represent ± 1 SE.

Table 4.1: Two-way ANOVA on WM behavioral performance

Source	Sum Sq.	df	Mean Sq.	F	P-value
repetition	67.82	11	6.165	0.29	0.9877
wmcond	524.89	2	262.444	28.9	$p < 0.001^{***}$
repetition:wmcond	199.74	22	9.079	1.94	0.0074^{**}

Two-way repeated measures ANOVA (Repetitions $\times$ WM conditions) of response errors across 17 participants revealed a significant main effect of condition ($F(2,16) = [28.9]$, $p < 0.001$).

Table 4.2: Paired t-test on WM behavioral performance

Pair	Mean 1	Stdev 1	Mean 2	Stdev 2	t	P-value
1 stim vs. 2 focused	4.17	1.90	5.25	1.66	-3.76	0.0017**
2 focused vs. 2 divided	5.25	1.66	6.44	2.13	-4.10	p<0.001***
1 stim vs. 2 divided	4.17	1.90	6.44	2.13	-7.12	p<0.001***

Paired t-tests revealed significant differences between all conditions: one stimuli versus two stimulus focused [one stimuli: $M = 4.17^\circ$, $SD = 1.90^\circ$; two stimulus focused: $M = 5.25^\circ$, $SD = 1.66^\circ$; $t(16) = -3.76$, $p = 0.0017$, two-tailed], two stimulus focused versus two stimulus divided [two stimulus focused: $M = 5.25^\circ$, $SD = 1.66^\circ$; two stimulus divided: $M = 6.44^\circ$, $SD = 2.13^\circ$; $t(16) = -4.10$, $p < 0.001$, two-tailed], and one stimuli versus two stimulus divided [one stimuli: $M = 4.17^\circ$, $SD = 1.90^\circ$; two stimulus divided: $M = 6.44^\circ$, $SD = 2.13^\circ$; $t(16) = -7.12$, $p < 0.001$, two-tailed].

4.1.2 Working Memory Model Parameter Analysis

For model selection, one stimulus condition data were fitted with mixture model and two stimuli conditions with swap model to replicate the previous study [15].

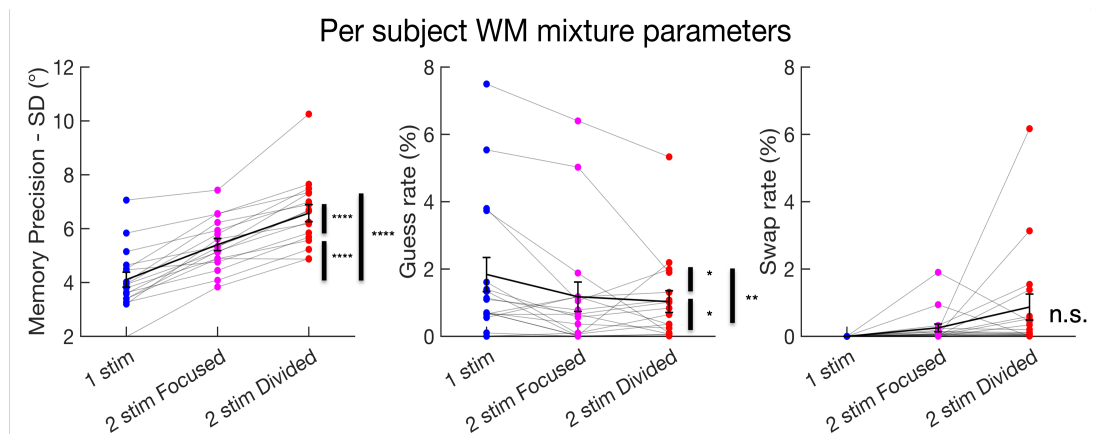


Figure 4.2: Working memory model parameters across conditions from every participants. Significant differences on mnemonic precision (SD) and guessing rate (g - converted to percentages) across conditions but not with swap rate (B - converted to percentages). Error bars represent ± 1 SE.

Table 4.3: One-way ANOVA on WM mixture (one stimuli condition) and swap (two stimulus conditions) SD parameter

Source	Sum Sq.	df	Mean Sq.	F	P-value
wmcond (SD)	51.832	2	25.9162	112.78	p<0.001***

Mnemonic precision (SD) worsen with increased load ($F(2,16) = [112.78]$, $p < 0.001$), while guessing rate exhibited smaller but significant differences ($F(2,16) = [8.04]$, $p = 0.017$). Swap rates remained negligible across conditions ($F(2,16) = [4.11]$, $p = 0.059$).

Table 4.4: Paired t-test on WM mixture and swap SD parameter

Pair	Mean 1	Stdev 1	Mean 2	Stdev 2	t	P-value
1 stim vs. 2 focused	4.11	1.15	5.42	0.95	-8.24	$p < 0.001^{***}$
2 focused vs. 2 divided	5.42	0.95	6.57	1.28	-7.57	$p < 0.001^{***}$
1 stim vs. 2 divided	4.11	1.15	6.57	1.28	-13.68	$p < 0.001^{***}$

Table 4.5: One-way ANOVA on WM mixture and swap g parameter

Source	Sum Sq.	df	Mean Sq.	F	P-value
wmcond (g)	0.00077	2	0.00038	8.04	0.0015**

4.2 Long-term Memory Task Results - Retrieval Block

4.2.1 Behavioral Performance

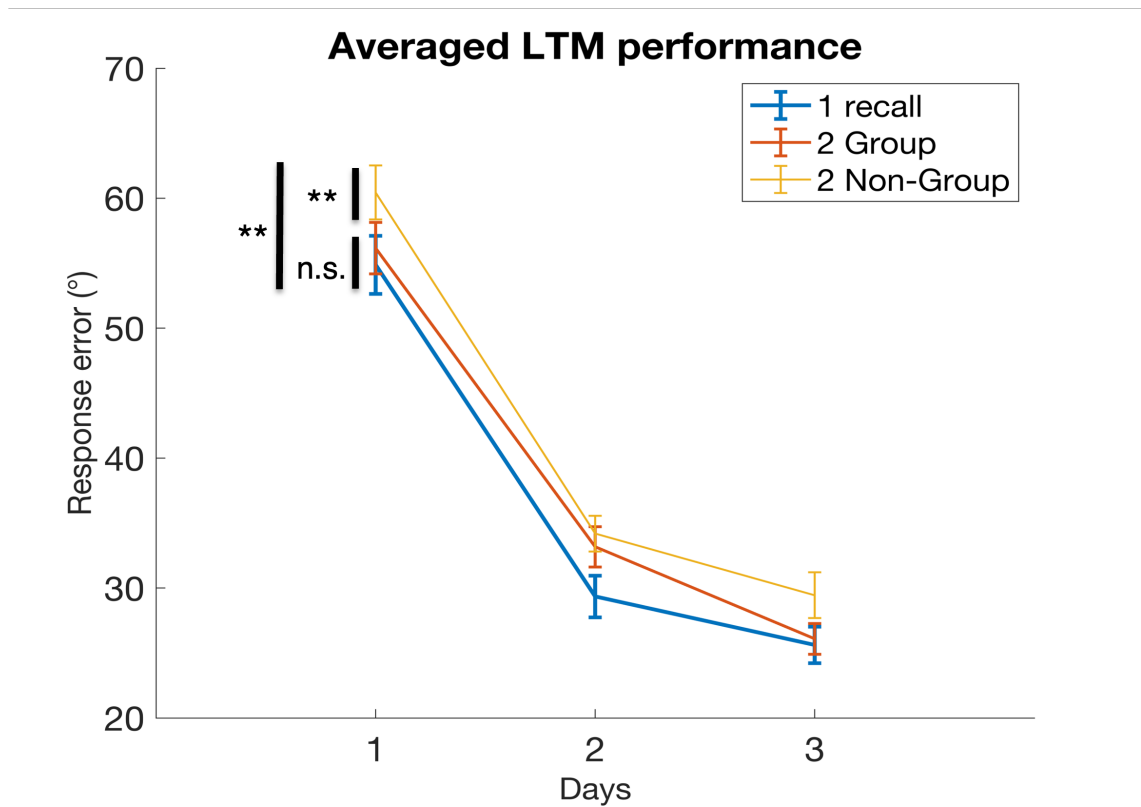


Figure 4.3: Averaged LTM task behavioral performance of all 17 participants. Learning curves showing significant improvement across days for all conditions. Error bars represent ± 1 SE.

Two-way repeated measures ANOVA (Session Days $\times$ LTM conditions) revealed significant main effects of both condition ($F(2, 16) = [8.45]$, $p = 0.0011$) and learning across days ($F(2, 16) = [83.31]$, $p < 0.001$).

Paired t-tests revealed significant differences only between two items grouped vs two items non-grouped [two items grouped: $M = 38.47^\circ$, $SD = 21.98^\circ$; two items non-grouped: $M = 41.35^\circ$, $SD = 24.25^\circ$; $t(16) = -3.87$, $p = 0.0013$, two-tailed], while there are no significant differences between one item and two items grouped [one item: $M = 36.61^\circ$, $SD = 23.69^\circ$; two items grouped: $M =$

Table 4.6: Paired t-test on WM mixture and swap g parameter

Pair	Mean 1	Stdev 1	Mean 2	Stdev 2	t	P-value
1 stim vs. 2 focused	0.02	0.02	0.01	0.01	3.26	0.0049**
2 focused vs. 2 divided	0.01	0.01	0.01	0.01	-2.14	0.047*
1 stim vs. 2 divided	0.02	0.02	0.01	0.01	2.43	0.027*

Table 4.7: One-way ANOVA on WM swap B parameter

Source	Sum Sq.	df	Mean Sq.	F	P-value
wmcond (B)	0.0001	1	0.0001	4.11	0.0597

38.47°, SD = 21.98°; $t(16) = -1.34$, $p = 0.1973$, two-tailed], signifying that attentional demands affect LTM retrieval performance.

4.2.2 LTM Model Parameter Analysis

Model comparison favored the mixture model over the swap model with both lowered AICc and BIC scores:

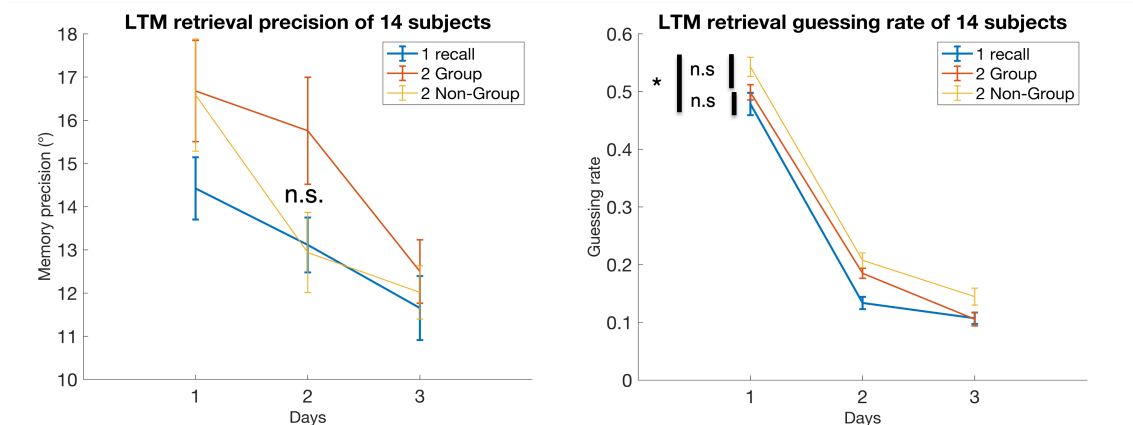


Figure 4.4: Averaged LTM model parameters of 14 participants. Mnemonic precision (SD) remains stable across conditions but improves with learning. Guessing rate (g) varies with condition and decreases with learning. Error bars represent ± 1 SE.

Mnemonic precision remained stable across conditions ($F(2,13) = 1.09$, $p = 0.3527$) but showed significant improvement across days. In contrast, guessing rate varied significantly with condition ($F(2,13) = 5.64$, $p = 0.0092$) and decreased with learning.

4.3 Discussion

Two-way ANOVA analysis of working memory performance revealed a significant main effect of condition ($F(2, 16) = [28.9]$, $p < 0.001$), with response errors progressively increasing from single to divided attention conditions. On model selection, one stimulus condition data were fitted with mixture model and two stimuli conditions with swap model to replicate the previous study [15]. The mixture model analysis demonstrated a significant decrease in mnemonic precision (SD) with increased load ($F(2, 16) = [112.78]$, $p < 0.001$) and significant differences in guessing rates across conditions ($F(2, 16) = [8.04]$, $p = 0.0015$). Swap rates in two-item conditions were negligible ($F(2, 16)$

Table 4.8: Paired t-test on WM swap B parameter

Pair	Mean 1	Stdev 1	Mean 2	Stdev 2	t	P-value
2 focused vs. 2 divided	0.0073	0.0052	0.010	0.010	-2.0268	0.0597

Table 4.9: Two-way ANOVA on LTM retrieval behavioral performance

Source	Sum Sq.	df	Mean Sq.	F	P-value
day	26428.1	2	13214.03	83.81	$p < 0.001^{***}$
ltmcond	583.2	2	291.6	8.45	0.0011^{**}
day:ltmcond	75.4	4	18.85	1.02	0.4057

= [4.11], $p = 0.059$). Overall, the trends aligned with the recent study in which supporting that WM is being modulated by attention [15], except that there were differences in guessing rates across conditions in which may cause from within-subject perceptual bias due to short cue duration of only 1250 ms [4].

For long-term memory retrieval, a two-way ANOVA revealed significant main effects of both condition ($F(2, 16) = [8.45]$, $p = 0.0011$) and learning across days ($F(2, 16) = [83.31]$, $p < 0.001$). There are significant response error differences that were averaged from all three days span between two items grouped and two items non-grouped conditions [two items grouped: $M = 38.4^\circ$, $SD = 21.9^\circ$; two items non-grouped: $M = 41.3^\circ$, $SD = 24.2^\circ$; $t(16) = -3.87$, $p = 0.0013$, two-tailed] but no differences between one item and two items grouped conditions [one item: $M = 36.6^\circ$, $SD = 23.6^\circ$; two items grouped: $M = 38.4^\circ$, $SD = 21.9^\circ$; $t(16) = -1.3452$, $p = 0.19$, two-tailed], which mean attentional demands have an effect on LTM retrieval performance. Model comparison favored the mixture model (see Table 4.11), with lower AICc (1020.8 vs 1022.3) and BIC (1034.1 vs 1040.1) scores. For mixture model analysis, LTM retrieval showed stable mnemonic precision across conditions ($F(2, 13) = [1.09]$, $p = 0.3527$) but significant changes in guessing rate ($F(2, 13) = [5.64]$, $p = 0.0092$) which signifies decision-making as an alternative neuromechanism that modulates LTM retrieval under varying attentional demands, also there are clear learning curves on both parameters across days, as shown in Fig. 4.4, confirming that the trends accurately describing LTM retrieval processes.

Table 4.10: Paired t-test on LTM retrieval behavioral performance

Pair	Mean 1	Stdev 1	Mean 2	Stdev 2	t	P-value
1 item vs. 2 grouped	36.61	23.69	38.47	21.98	-1.34	0.1973
2 grouped vs. 2 non-grouped	38.47	21.98	41.35	24.25	-3.87	0.0013**
1 item vs. 2 non-grouped	36.61	23.69	41.35	24.25	-3.75	0.0017**

Table 4.11: Model comparison metrics for LTM task computed from sum of 14 participant scores on all 3 conditions

Metric	Mixture Model	Swap Model
AICc	1020.8	1022.3
BIC	1034.1	1040.1

Table 4.12: Two-way ANOVA on LTM retrieval mixture SD parameter

Source	Sum Sq.	df	Mean Sq.	F	P-value
day	309.2	2	154.601	18.14	p<0.001***
ltmcond	77.83	2	38.914	1.09	0.3527
day:ltmcond	42.58	4	10.644	1.12	0.3584

Table 4.13: Paired t-test on LTM retrieval mixture SD parameter

Pair	Mean 1	Stdev 1	Mean 2	Stdev 2	t	P-value
1 item vs. 2 grouped	13.06	2.92	14.97	6.53	-1.39	0.18
2 grouped vs. 2 non-grouped	14.97	6.53	13.84	5.27	0.74	0.47
1 item vs. 2 non-grouped	13.06	2.92	13.84	5.27	-0.82	0.42

Table 4.14: Two-way ANOVA on LTM retrieval mixture g parameter

Source	Sum Sq.	df	Mean Sq.	F	P-value
day	3.68391	2	1.84195	98.08	p<0.001***
ltmcond	0.07311	2	0.03655	5.64	0.0092**
day:ltmcond	0.01117	4	0.00279	1.13	0.352

Table 4.15: Paired t-test on LTM retrieval mixture g parameter

Pair	Mean 1	Stdev 1	Mean 2	Stdev 2	t	P-value
1 item vs. 2 grouped	0.23	0.12	0.26	0.11	-1.77	0.098
2 grouped vs. 2 non-grouped	0.26	0.11	0.29	0.16	-1.93	0.074
1 item vs. 2 non-grouped	0.23	0.12	0.29	0.16	-2.85	0.013*

Chapter 5

Conclusions

This chapter summarizes key findings from our investigation of neuromechanisms underlying long-term memory retrieval and discusses their implications for understanding human memory systems.

5.1 Summary

Through comparison of working memory and long-term memory retrieval processes, our study revealed several key findings regarding the distinct mechanisms involved. In working memory experiments, we found that attention strongly modulates memory precision - performance decreased systematically as attentional load increased, with mnemonic precision (SD) showing significant degradation across conditions. This pattern aligns with previous work suggesting attention as the primary mechanism in working memory.

The long-term memory experiments, however, revealed a different pattern. On retrieval block, mnemonic precision remained stable across retrieval conditions, while guessing rates showed variation with task demands. This dissociation suggests that long-term memory retrieval relies more heavily on decision-making processes rather than attentional mechanisms.

Over the three days of testing, we observed significant learning effects, with performance improving from Day 1 to Day 3. Importantly, the pattern of stable precision with varying guessing rates persisted across all days, suggesting this reflects a fundamental characteristic of long-term memory retrieval rather than a transient effect.

5.2 Contributions

This research makes several significant contributions to our understanding of human memory systems. First, it provides the supporting evidence for distinct neuromechanisms in working memory versus long-term memory retrieval. The opposing patterns in memory precision and guessing rates challenge prevailing models that assume similar attentional mechanisms operate across all memory processes [1] [3].

The finding that long-term memory retrieval appears to rely more on decision-making processes than attention has important practical implications. It suggests that interventions aimed at improving long-term memory might be more effective if they target decision-making strategies rather than attentional control. This could inform the development of more targeted approaches for memory enhancement and rehabilitation [9].

5.3 Future Works

Several promising directions emerge from this research. First, the dissociation between working memory and long-term memory mechanisms could be further investigated using neuro-

electrophysiology techniques. Tracking neural activity during memory retrieval could help identify the specific brain networks involved in these distinct processes and validate our behavioral findings at the neuronal population level.

Investigation of clinical populations with memory impairments represents another important direction. If long-term memory retrieval indeed relies primarily on decision-making mechanisms, this suggests novel therapeutic approaches for conditions affecting memory. Future studies could examine whether training decision-making processes might provide more effective rehabilitation than traditional attention-based interventions.

The computational modeling approach could be extended to more complex real-world memory tasks. While our study used simple location memory to enable precise measurement, examining how these mechanisms operate in more naturalistic settings would be valuable. This might include studying memory for complex scenes, autobiographical events, or cross-modal associations.

Through these future directions, we hope to build on the foundation established by this research to develop a more complete understanding of human memory systems and their underlying mechanisms. This knowledge will be crucial for both advancing theoretical models and developing more effective interventions for memory enhancement and rehabilitation.

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Appendix

Appendix A

Experimental Documents

ขั้นตอนการเข้าร่วมวิจัย: หากท่านยินดีที่จะเข้าร่วมการวิจัยนี้

ท่านจะต้องแจ้งนัดวันที่ท่านสะดวกเข้าร่วมการวิจัยทางอีเมลอย่างน้อยสองวันก่อนถึงวันที่นัด การวิจัยจะใช้เกมบนคอมพิวเตอร์ และใช้เวลาราวๆ 1 ถึง 2 ชม. โดยจะมีช่วงเวลาให้พักระหว่างการเล่นทุกๆ 10 นาที โดยท่านจะสามารถพักได้ประมาณ 3-5 นาที ท่านจะมองสิ่งเร้าที่แสดงบนหน้าจอเรื่อยๆ

โดยให้ท่านฟังสมาริไปที่สิ่งเร้า พยายามไม่เลื่อนสายตาออกจากจุดเพ่งเล็งกลางหน้าจอและตัดสินใจเกี่ยวกับสิ่งเร้าโดยใช้การคลิกเมาส์บนบริเวณที่กำหนด ท่านจะได้รับผลลัพธ์จากการคลิก เช่นค่าความคลาดเคลื่อนของตำแหน่งที่ตอบ ขณะที่เล่นเกมคอมพิวเตอร์

เราจะติดตามความสามารถในการเล่นเกมนของท่าน โดยคำนวณค่าความคลาดเคลื่อนของตำแหน่งที่ตอบ และความเร็วในการตอบ

เกมที่ 1 Working Memory:

ในเกมนี้ ท่านจะต้องจดจำตำแหน่งของจุดสีที่ปรากฏขึ้นบนหน้าจอ ซึ่งมีโอกาสที่จะมีจุดสีเพียงจุดเดียว หรือมีสองจุดสีปรากฏขึ้นมาพร้อมกัน

จากนั้นท่านจะต้องตอบกลับตำแหน่งของจุดสีที่จะถาม ซึ่งจะถามเพียงสีเดียวโดยจะปรากฏบนจุดเพ่งเล็งกลางหน้าจอ

ให้ท่านตอบกลับโดยใช้เมาส์คลิกบนวงแหวน ณ ตำแหน่งที่จดจำมาก่อนหน้านี้ให้ได้แม่นยำมากที่สุด จากนั้นจะแสดงผลค่าความคลาดเคลื่อนของตำแหน่งที่ตอบ และแสดงตำแหน่งที่แท้จริงของจุดสีที่ถาม

เกมที่ 2 Study:

ในเกมนี้ ท่านจะต้องจดจำความสัมพันธ์ระหว่างรูปภาพและตำแหน่งของจุดสีดำที่ปรากฏขึ้นบนหน้าจอ

จากนั้นท่านจะต้องตอบกลับตำแหน่งของจุดสีดำ

ให้ท่านตอบกลับโดยใช้เมาส์คลิกบนวงแหวน ณ ตำแหน่งที่จดจำมาก่อนหน้านี้ให้ได้
แม่นยำมากที่สุด จากนั้นจะแสดงผลค่าความคลาดเคลื่อนของตำแหน่งที่ตอบ และ
แสดงตำแหน่งที่แท้จริงของจุดสีดำ
จะมีทั้งหมด 162 ความสัมพันธ์ของรูปภาพและตำแหน่งที่ท่านจะต้องจดจำในเกมนี้

เกมที่ 3 Retrieval:

ในเกมนี้ ท่านจะต้องใช้ความรู้ของความสัมพันธ์ทั้งหมด 162 รูปภาพและตำแหน่ง
ที่ท่านได้จดจำมาจากเกม Study โดยจะมีรูปภาพสีปรากฏขึ้น ซึ่งมีโอกาสที่จะมี
รูปภาพสีเพียงรูปเดียวโดยในด้านตรงข้ามจะเป็นเพียงรูปแตกสลายสีดำที่ไม่ต้อง
สนใจ หรือมีสองรูปภาพสีปรากฏขึ้นมาพร้อมกัน
ในบางครั้ง รูปภาพสีที่ปรากฏขึ้นจะไม่ถูกถาม และจะข้ามไปยังรูปภาพสีชุดต่อไป
แทน

จากนั้นท่านจะต้องตอบกลับตำแหน่งที่สัมพันธ์กับรูปภาพสีที่จะถาม โดยจะมีสีของ
รูปภาพสีที่จะถาม ซึ่งจะถามเพียงสีเดียวโดยจะปรากฏบนจุดเพ่งเล็งกลางหน้าจอ
ให้ท่านตอบกลับโดยใช้เมาส์คลิกบนวงแหวนสีดำ ณ ตำแหน่งที่จดจำมาก่อนหน้านี้
ให้ได้แม่นยำมากที่สุด จากนั้นจะแสดงผลค่าความคลาดเคลื่อนของตำแหน่งที่ตอบ
และแสดงตำแหน่งที่แท้จริงของตำแหน่งที่สัมพันธ์กับรูปภาพสีที่ถาม

เกมที่ 4 Retrieval individual:

ในเกมนี้ ท่านจะต้องใช้ความรู้ของความสัมพันธ์ทั้งหมด 162 รูปภาพและตำแหน่ง
ที่ท่านได้จดจำมาจากเกม Study โดยจะมีรูปภาพสีเพียงรูปเดียวปรากฏขึ้น โดยใน
ด้านตรงข้ามจะเป็นเพียงรูปแตกสลายสีดำที่ไม่ต้องสนใจ

จากนั้นท่านจะต้องตอบกลับตำแหน่งที่สัมพันธ์กับรูปภาพสีที่จะถาม โดยจะมีสีของ
รูปภาพสีที่จะถาม ซึ่งจะถามเพียงสีเดียวโดยจะปรากฏบนจุดเพ่งเล็งกลางหน้าจอ

ให้ท่านตอบกลับโดยใช้เมาส์คลิกบนวงแหวนสีดำ ณ ตำแหน่งที่จดจำมาก่อนหน้านี้
ให้ได้แม่นยำมากที่สุด จากนั้นจะแสดงผลค่าความคลาดเคลื่อนของตำแหน่งที่ตอบ
และแสดงตำแหน่งที่แท้จริงของตำแหน่งที่สัมพันธ์กับรูปภาพสีที่ถาม

Figure A.1: Experimental protocol in verbatim for informing participants.